

Demographic Patterns of Melanoma: Comparing Incidence and Survival Among Black and White Populations

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TODO: The limit of our report is 5 pages. Use the ggplot2 library in r to display the results. (Just copy the images from the R code; no need to use all of the code in this report.)

Abstract

TODO: Summarize the project here.

Introduction

Melanoma is the most aggressive form of skin cancer, originating from melanocytes in the epidermis. Incidence has risen sharply over recent decades. This increase in incidence is particularly in fair-skinned populations which makes being fair-skinned a major risk factor (Culp MB et al., 2019). Epidemiological data show that melanoma is more common in individuals with lighter skin, a history of sunburns especially before the age of 20, numerous or atypical nevi, a family history of melanoma, being male and older age (Culp MB et al., 2019). Other risk factors include exposure to high levels of ultraviolet radiation, immunosuppression, indoor tanning, and genetic mutations such as CDKN2A and BRAF. Based on lifetime risk statistics, White individuals have a melanoma risk ratio of approximately 26 to 1 compared to Black individuals. This is calculated using the American Cancer Society's lifetime risk figures of 1 in 38 for White people and 1 in 1,000 for Black people (Culp MB et al., 2019). However, few studies of melanoma were conducted among non-Hispanic black populations and although black men and women in the United States are at lower risk of melanoma than white men and women, they often have melanoma diagnosed at a later stage and have poorer survival as a result (Culp MB et al., 2019).

In this project, we will use data from the Surveillance, Epidemiology, and End Results (SEER) dataset. The SEER dataset is a federally funded cancer registry, established by the National Cancer Institute. It compiles data from population-based cancer registries covering nearly half of the U.S. population and provides comprehensive information on patient demographics (age, sex, race, ethnicity), tumor characteristics (site, stage, grade, size), initial treatment data, and survival outcomes. SEER data are organized into cohorts (e.g., SEER-9, SEER-13, SEER-18) that reflect the expansion of registries over time, ensuring longitudinal tracking of cancer trends (National Cancer Institute, 2025).

The aim of this research project is to answer the following question: **What are the differences in melanoma cancer incidence and survival rates between Black and White populations?** By leveraging the robust data collection of the SEER program's population-based registries, we will systematically and statistically identify disparities and emerging trends in melanoma outcomes. Our analysis will focus on incidence rates and survival patterns across disease stages, with particular attention to race as a critical demographic factor. By integrating these elements and applying advanced statistical methods including, but not limited to, joinpoint regression models

and survival modeling, we will generate actionable insights into inequities in melanoma outcomes. Ultimately, this work will hopefully be used to inform evidence-based strategies for targeted interventions, contributing to improved health equity and cancer control.

Methods

TODO: What did we use to run our code?

Results

TODO: What did our code output?

Conclusion

TODO: What are our takeaways?

References

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Supplementary Information

Group Member Contributions:

- **Violette Bonvallet:** Contributed to background analysis
- **Andre Gala-Garza:** Initiated and maintained the GitHub repository; Reformatted code for style and efficiency
- **Lynle Hayes:** Designed the initial project proposal; Created initial code base

GitHub Repository: <https://github.com/AndreGalaGarza/melanomaBD>
(<https://github.com/AndreGalaGarza/melanomaBD>)